

A B C D E

③

M ₁	9	14	19	15	8
M ₂	7	17	20	19	0
M ₃	9	18	21	18	0
M ₄	10	12	18 18	19	0
M ₅	10	15	21	16	0

2	—	2	—	1	—	<u>0</u>	—	0
<u>0</u>	—	5	—	2	—	4	—	0
2		6		3		3		0
3	—	<u>0</u>	—	0	—	4	—	0
3		3		3		1		<u>0</u>

2	2	1	0	1	2	2	1	0	1
0	5	2	4	1	0	5	2	4	1
1	5	2	2	0	1	5	2	2	0
3	0	0	4	1	3	0	0	4	1
2	2	2	0	0	2	2	2	0	0

1	1	<u>0</u>	0	1						
<u>0</u>	5	2	5	2	=	19	+	7	+	0
0	4	7	2	<u>0</u>	+	12	+	16		
3	<u>0</u>	0	5	2						
2	2	2	<u>0</u>	0						

= 54 hours.

$$\begin{array}{r} 2-3-0 \\ 4 \quad 1 \\ \hline 4 \quad 4 \quad 6 \end{array}$$

$$+ 3 + 2$$

m.

$$\begin{array}{r} 3 \quad 2 \\ \times 7 \\ \hline 21 \\ 14 \\ \hline 21 \end{array}$$

	A	B	C	D
S ₁	40	50	60	65
S ₂	30	38	46	49
S ₃	25	33	41	43
S ₄	39	45	51	59

$$\begin{array}{r} 10 \quad 4 \quad 8 \quad 7 \\ 0 \quad 2 \quad 4 \quad 0 \\ 0 \quad 2 \quad 4 \quad 0 \\ 0-0-0-2 \end{array} \quad \begin{array}{r} 0 \\ 0 \\ 0 \\ 2 \end{array} \quad \begin{array}{r} 2 \\ 2 \\ 2 \\ 0 \end{array} \quad \begin{array}{r} 6 \\ 2 \\ 2 \\ 10 \end{array} \quad \begin{array}{r} 7 \\ 0 \\ 0 \\ 24 \end{array}$$

$$= 40 + 48 + 25 + 51$$

$$= 172 \text{ miles.}$$

$$\begin{array}{r} 1 \quad 0 \quad 10 \quad 0 \\ 10 \quad 1 \quad 6 \quad 0 \\ 12 \quad 0 \quad 2 \quad 0 \\ 13 \quad 3 \quad 6 \quad 0 \end{array} = 11 + 8 + 12 + 0$$

$$= 31 \text{ hrs}$$

(4)

②

	S_1	S_2	S_3	S_4
M_1	31	32	33	34
M_2	14	5	8	7
M_3	2	12	6	5
M_4	7	8	3	9
	2	4	6	10

9	0	3	2	19-0-3-0
0	10	4	3	10 10 4 1
4	5	0	6	14-5-19-4
0	2	4	8	0 2 4 6

$10 \boxed{0} \boxed{3} \boxed{0}$
 $1 \ 9 \ \boxed{3} \boxed{0} = 5 + 5 + 3 + 2$
 $5 \ 5 \ \boxed{0} \ 4 = 15 \text{ hrs.}$
 $\boxed{0} \ 1 \ 3 \ 5$

③

	I	II	III	IV
JA	9	12	11	0
JB	8	13	17	0
Jc	20	12	13	0
JD	21	15	17	0

0	3	1
0	5	6
8	0	0
6	0	2

0	3	2
0	5	7
8	0	1
6	0	2

⑤

S_1
 S_2
 S_3
 S_4

0
0
0
0

③

④

Sample 1

①

	X	Y	Z
A	11	16	21
B	20	13	17
C	13	15	12

0	5	10
7	0	4
1	3	0

$$= 11 + 13 + 12$$

$$= 36 \text{ hrs}$$

∴ Minimum total processing Time = 36 hours

②

	J ₁	J ₂	J ₃	J ₄
M ₁	14	5	8	7
M ₂	2	12	6	5
M ₃	7	8	3	9
M ₄	2	4	6	10

9	0	3	2
0	10	4	3
4	5	0	6
0	2	4	8

11	0	3	2
0	8	2	1
6	5	0	6
0	0	2	6

9	0	3	0
0	10	4	1
4	5	0	4
0	2	4	6

11	0	3	0
0	8	2	1
6	5	0	4
0	0	2	4

$$= 7 + 2 + 3 + 4$$

$$= 16 \text{ hrs}$$